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Buck converter including a bootstrap capacitor and an operating method thereof

Abstract

A buck converter including: a first switching element connecting an input voltage node to a switching node; a second switching element connecting the switching node to a ground node; a low drop out (LDO) voltage regulator connected to the input voltage node; a pulse width modulation (PWM) controller configured to receive a mode signal indicating an operation mode and toggle, in response to the mode signal, a first control signal for turning on the first switching element or a second control signal for turning on the second switching element; a bootstrap charger configured to receive the mode signal and toggle, in response to the mode signal, a third control signal for turning on the second switching element; and a bootstrap capacitor connecting an output node of the LDO voltage regulator to the switching node.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0110313, filed on Aug. 20, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

(2) The inventive concept relates to a buck converter, and more particularly, to a buck converter including a bootstrap capacitor and an operating method thereof.

DISCUSSION OF RELATED ART

(3) As electronic technologies continue to develop, various types of electronic device's have been used. One example is a mobile device such as a smartphone, laptop or wearable. A mobile electronic device may be driven using a battery device included therein. As power consumption of electronic devices has increased, battery capacities have increased, and accordingly, batteries may be charged at various speeds.

SUMMARY

(4) The inventive concept provides a device and method to maintain a voltage level of a bootstrap capacitor based on periodic charging of the bootstrap capacitor.

(5) According to an example embodiment of the inventive concept, a buck converter includes: a first switching element connecting an input voltage node to a switching node; a second switching element connecting the switching node to a ground node; a low drop out (LDO) voltage regulator connected to the input voltage node; a pulse width modulation (PWM) controller configured to receive a mode signal indicating an operation mode and toggle, in response to the mode signal, a first control signal for turning on the first switching element or a second control signal for turning on the second switching element; a bootstrap charger configured to receive the mode signal and toggle, in response to the mode signal, a third control signal for turning on the second switching element; and a bootstrap capacitor connecting an output node of the LDO voltage regulator to the switching node.

(6) According to an example embodiment of the inventive concept, an operating method of a buck converter includes: receiving a mode signal and identifying an operation mode; toggling, based on the operation mode, a first control signal for turning on a first switching element and a second control signal for turning on a second switching element; and toggling a third control signal for turning on the second switching element.

(7) According to an example embodiment of the inventive concept, a buck converter includes: a first switching element connecting an input voltage node to a switching node; a second switching element connecting the switching node to a ground node; a pulse width modulation (PWM) controller configured to receive a mode signal indicating an operation mode and toggle, in response to the mode signal, a first control signal for turning on the first switching element and a second control signal for turning on the second switching element; a bootstrap charger configured to receive the mode signal and toggle, in response to the mode signal, a third control signal for turning on the second switching element; and a bootstrap capacitor connecting the first switching element to the switching node.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the inventive concept will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:

- (2) FIG. 1 is a block diagram illustrating an electronic device and an external power source according to an example embodiment of the inventive concept;
- (3) FIG. 2 is a detailed block diagram illustrating a charging integrated circuit according to an example embodiment of the inventive concept;
- (4) FIG. 3 is a circuit diagram illustrating a buck converter according to an example embodiment of the inventive concept;
- (5) FIG. 4 is a timing diagram illustrating a pulse skip mode according to an example embodiment of the inventive concept;
- (6) FIG. 5 is a timing diagram illustrating an asynchronous mode according to an example embodiment of the inventive concept;
- (7) FIG. 6 is a circuit diagram illustrating a buck converter including a bootstrap capacitor, according to a comparative example;
- (8) FIG. 7A is a timing diagram illustrating the pulse skip mode according to a comparative example;
- (9) FIG. 7B is a timing diagram illustrating the pulse skip mode according to a comparative example; and
- (10) FIG. 8 is a timing diagram illustrating the asynchronous mode according to a comparative example.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(11) Hereinafter, embodiments of the inventive concept are described in detail with reference to the accompanying drawings.

(12) FIG. 1 is a block diagram illustrating an electronic device and an external power source according to an example embodiment of the inventive concept.

(13) Referring to FIG. 1, an electronic device **100** and an external power source **200** are shown. To charge a battery **130** included in the electronic device **100**, the electronic device **100** and the external power source **200** may be connected to each other in a wired or wireless manner. In other words, the battery **130** may be charged using a wired or wireless charging technique.

(14) In various embodiments of the inventive concept, the electronic device **100** may include various kinds of electronic equipment which a user may carry. For example, the electronic device **100** may be a mobile device such as a smartphone, a tablet personal computer (PC), a cellular phone, a personal digital assistant (PDA), a laptop computer, a wearable device, a global positioning system (GPS) device, an e-book terminal, a digital broadcasting terminal, a Moving Picture Experts Group (MPEG) Audio Layer-3 (MP3) player, or a digital camera. As another example, the electronic device **100** may be an electric vehicle.

(15) The electronic device **100** may include a charging integrated circuit (IC) **110**, and the charging IC **110** may be referred as a “battery charger”. For example, the charging IC **110** may be implemented by an IC chip and mounted on a printed circuit board (PCB). The charging IC **110** may receive power from the external power source **200**, charge the battery **130** by providing the power to the battery **130**, and control the electronic device **100** to perform various functions by providing the power to a system load **120**. A particular configuration of the charging IC **110** is described below with reference to FIG. 2.

(16) In addition, the electronic device **100** may include the battery **130**. The battery **130** may include at least one battery cell. For example, the battery **130** may correspond to a multi-cell battery including a plurality of battery cells connected in series to each other. As another example, the battery **130** may correspond to a single-cell battery including a single battery cell. The battery **130** may receive power through the charging IC **110** when the electronic device **100** is connected to the external power source **200**. When the electronic device **100** is not connected to the external power source **200**, the battery **130** may provide power to the system load **120** to control the electronic device **100** to perform various functions.

(17) In addition, the electronic device **100** may include the system load **120**. The system load **120**

may include the components remaining by excluding the charging IC **110** and the battery **130** from the electronic device **100**. For example, the system load **120** may include a display, an application processor, a communication processor, a speaker, a memory, and the like. In other words, the system load **120** may include chips, modules, operation blocks, function blocks, and intellectual property (IP) blocks included in the electronic device **100**. The system load **120** may receive power from the external power source **200** or the battery **130** and provide various functions to the user. For example, a monitor in the system load **120** may provide visual awareness to the user by displaying an object through the display, the communication processor may transmit and receive data by exchanging a wireless signal with an external device, and the application processor may perform various calculations.

(18) In addition, the electronic device **100** may include a receptacle interface **140**. The receptacle interface **140** may connect the electronic device **100** to the external power source **200** via a universal serial bus (USB) cable. In an embodiment of the inventive concept, the receptacle interface **140** may correspond to a USB type-C interface, and the USB cable may correspond to a USB type-C cable. The USB type-C interface may be implemented based on the definition of USB 2.0 or USB 3.1. The receptacle interface **140** may include a plurality of pins. The plurality of pins may include pins for power supply, pins for data transmission, and a configuration channel (CC) pin.

(19) The external power source **200** may supply power to the electronic device **100**. According to various embodiments of the inventive concept, the external power source **200** may include a travel adaptor (TA) **210** and a wireless charger **220**. The wireless charger **220** may charge the electronic device **100** by wirelessly transmitting power through the air instead of supplying power through a wire. According to various embodiments of the inventive concept, the wireless charger **220** may transmit power based on various wireless charging schemes such as a magnetic induction scheme, a magnetic resonance scheme, an electromagnetic induction scheme, and a non-radiative wireless charging (e.g., WiTricity) scheme.

(20) The TA **210** may supply power through a wire connected to the charging IC **110** in the electronic device **100**. The TA **210** may convert household electric power, e.g., alternating current (AC) 110 V to 220 V, or power supplied from another power supply (e.g., a computer) into direct current (DC) power required to charge the battery **130**, and provide the DC power to the electronic device **100**. According to various embodiments of the inventive concept, the receptacle interface **140** may be electrically connected to an output terminal of an auxiliary battery. According to various embodiments of the inventive concept, the TA **210** may support direct charging. As to be described below, an application processor (**121**, see FIG. 2) or a direct charger (**113**, see FIG. 2) may identify whether the TA **210** supports direct charging, by using a resistance value of a CC pin (**141**, see FIG. 2).

(21) FIG. 2 is a detailed block diagram illustrating a charging IC according to an example embodiment of the inventive concept.

(22) Referring to FIG. 2, the TA **210** and the electronic device **100** including the charging IC **110** are disclosed. Although FIG. 2 shows that the electronic device **100** is connected to the TA **210**, the example embodiment is not limited thereto. For example, the inventive concept is also applicable to an embodiment in which the electronic device **100** is connected to the wireless charger **220** in the external power source **200**. In addition, although only the application processor **121** is shown as an example of the system load **120**, the inventive concept is not limited thereto.

(23) According to various embodiments of the inventive concept, the charging IC **110** may include a control circuit **112**, the direct charger **113**, a switching charger **114**, an inrush current limiter (ICL) **111**, a fuel gauge **116**, and a power meter **115**.

(24) The ICL **111** may adjust a magnitude of an input current. For example, when the battery **130** is being charged and the system load **120** (in this case, the application processor **121**) is also operating at the same time, a magnitude of an input current input to the charging IC **110** may

correspond to a sum of a charging current for charging the battery **130** and a load current provided to the system load **120**. An increase in the magnitude of the input current may cause an overload of the charging IC **110** and risks such as fire may occur due to the overload. Therefore, the ICL **111** may control the magnitude of the input current so as not to exceed a certain magnitude. The ICL **111** may include an internal transistor resistance. When the magnitude of the input current exceeds the certain magnitude, a magnitude of the internal transistor resistance may be increased in response to a control signal received from the control circuit **112**. Because a magnitude of an external power source voltage is constant, the magnitude of the input current may be reduced by increasing the magnitude of the internal transistor resistance.

(25) The control circuit **112** may control a general operation of the charging IC **110**. The control circuit **112** may communicate with the TA **210** via the receptacle interface **140**. For example, the control circuit **112** may control an output voltage and an output current of the TA **210** by transmitting control information to the TA **210**. As another example, the control circuit **112** may identify the connected TA **210** through the CC pin **141**.

(26) The switching charger **114** includes a buck converter and may adjust a charging current by stepping down an input voltage and adjusting a cycle of the buck counter. Because the switching charger **114** includes an inductor, switching loss and conduction loss due to a resistance component of the inductor cannot be removed. Therefore, a charging efficiency of the switching charger **114** may be lower than a charging efficiency of the direct charger **113**. A particular operation of the buck converter is described below with reference to FIG. 3.

(27) The direct charger **113** may directly provide the input voltage to the battery **130** via a cap (or capacitive) divider. Because the direct charger **113** includes only a transistor and a capacitor, the switching loss and the conduction loss due to the resistance component of the inductor may be saved. The direct charger **113** may be suitable for a highly efficient charging system because the input voltage is directly provided to a node of the battery **130** via the cap divider.

(28) The fuel gauge **116** may sense the battery **130**. For example, the fuel gauge **116** may sense a state of charge (SoC) of the battery **130**, a charging voltage, a charging current, and a battery temperature. The SoC is a ratio of a current capacity of the battery **130** to a maximum capacity of the battery **130** and may be defined by a unit of percentage (%).

(29) The fuel gauge **116** may include an analog to digital (ADC) converter. The fuel gauge **116** may convert analog information of a sensed voltage, current, and temperature into digital information by using the ADC converter and provide the digital information to the direct charger **113** and/or the application processor **121**. The fuel gauge **116** may be implemented as a part of an IC chip and mounted on a PCB. However, the inventive concept is not limited thereto, and the fuel gauge **116** and the charging IC **110** may be implemented by different ICs or IC chips, respectively.

(30) The power meter **115** may acquire information about input and output voltages and currents. The power meter **115** may sense magnitudes of a voltage and a current input to the charging IC **110** and magnitudes of a voltage and a current provided to the system load **120**.

(31) The power meter **115** may include an ADC converter, convert analog information of a sensed voltage, current, and power into digital information by using the ADC converter, and provide the digital information to the application processor **121** and/or the direct charger **113**. The power meter **115** may be implemented as a part of an IC chip and mounted on a PCB. However, the inventive concept is not limited thereto, and the power meter **115** and the charging IC **110** may be implemented by different ICs or IC chips, respectively.

(32) A mode change from the direct charger **113** to the switching charger **114** may be performed based on information received from the fuel gauge **116** and the power meter **115**. The charging IC **110** may further include a circuit or a block supporting at least one of various functions such as an under-voltage lockout (UVLO) function, an over-current protection (OCP) function, an over-voltage protection (OVP) function, a soft-start function for reducing an inrush current, a foldback current limit function, a hiccup mode function for short circuit protection, and an over-temperature

protection (OTP) function so as to properly operate in a power saving condition.

(33) FIG. 3 is a circuit diagram illustrating a buck converter **300** according to an example embodiment of the inventive concept.

(34) Referring to FIG. 3, the buck converter **300** may include a first power metal-oxide-semiconductor field-effect transistor (MOSFET) **310**, a second power MOSFET **320**, a bootstrap capacitor (CBTS) **330**, a pulse width modulation (PWM) controller **340**, a cyclic bootstrap charger **350**, and a low drop out (LDO) voltage regulator **360**. The buck converter **300** may further include an inductor connected between a switching node LX and an output voltage node VOUT, a capacitor may be provided between the output voltage node VOUT and a ground node and an output current I_OUT may be provided on a path between the output voltage node VOUT and the ground node. In an example embodiment, the first power MOSFET **310** may connect an input voltage node VIN to a switching node LX, the second power MOSFET **320** may connect the switching node LX to a ground node, the LDO voltage regulator **360** may be connected to the input voltage node VIN, and the bootstrap capacitor **330** may connect an output node of the LDO voltage regulator **360** to the switching node LX. In an example embodiment, the buck converter **300** may not include the LDO voltage regulator **360**, and the bootstrap capacitor **330** may connect the first power MOSFET **310** to the switching node LX.

(35) According to an embodiment of the inventive concept, the PWM controller **340** may output a first control signal PWM_HS and a second control signal PWM_LS. A first gate drive signal GD_HS may be a control signal for turning on/off the first power MOSFET **310**. For example, the first gate drive signal GD_HS may transit between 0 and 1 based on the first control signal PWM_HS. A second gate drive signal GD_LS may be a control signal for turning on/off the second power MOSFET **320**. For example, the second gate drive signal GD_LS may transit between 0 and 1 based on the second control signal PWM_LS or a third control signal BTS_CHG. The second gate drive signal GD_LS may be logic high when either the second control signal PWM_LS and the third control signal BTS_CHG transit to logic high. The PWM controller **340** may adjust an output ratio of the first control signal PWM_HS to the second control signal PWM_LS in response to a mode signal SIG_MODE indicating an operation mode. The first power MOSFET **310** may be referred as an upper switching element, and the second power MOSFET **320** may be referred as a lower switching element. The first power MOSFET **310** may also be referred as a first switching element, and the second power MOSFET **320** may be referred as a second switching element.

(36) For example, the PWM controller **340** may receive a control signal indicating that the operation mode is a pulse skip mode. The pulse skip mode may correspond to a mode of skipping toggling of the first control signal PWM_HS applied to the first power MOSFET **310**. In other words, when entering the pulse skip mode, the first control signal PWM_HS may be maintained "logic low". In other words, when entering the pulse skip mode, the first control signal PWM_HS may not be toggled.

(37) As another example, the PWM controller **340** may receive a control signal indicating that the operation mode is an asynchronous mode. The asynchronous mode may be a mode applied so that the first control signal PWM_HS and the second control signal PWM_LS do not alternate. In other words, when entering the asynchronous mode, the first power MOSFET **310** may operate to periodically repeat turning-on and turning-off, while the second power MOSFET **320** maintains being turned-off during the asynchronous mode. In another embodiment of the inventive concept, when entering the asynchronous mode, the second power MOSFET **320** may operate to periodically repeat turning-on and turning-off, while the first power MOSFET **310** maintains being turned-off during the asynchronous mode.

(38) The bootstrap capacitor **330** may boost a voltage to be input to a gate of the first power MOSFET **310**. The bootstrap capacitor **330** may further include a diode, wherein the bootstrap capacitor **330** may be instantaneously charged with a voltage to apply a voltage sufficiently greater than a gate-source voltage drop to the gate of the first power MOSFET **310**, thereby operating the

first power MOSFET **310**. According to an embodiment of the inventive concept, the bootstrap capacitor **330** may be connected between a switching node LX and an input voltage node VIN. (39) The cyclic bootstrap charger **350** may periodically provide the third control signal BTS_CHG to the second power MOSFET **320**. The second power MOSFET **320** may be connected to a ground node. The cyclic bootstrap charger **350** may be activated in the asynchronous mode or the pulse skip mode of the operation mode. In the asynchronous mode and the pulse skip mode, because the second control signal PWM_LS for the first power MOSFET **310** is always inactivated, a voltage level VBTS of the bootstrap capacitor **330** may decrease due to a leakage current generated by the bootstrap capacitor **330**. To prevent the decrease in the voltage level of the bootstrap capacitor **330**, the cyclic bootstrap charger **350** may generate the third control signal BTS_CHG for turning on the second power MOSFET **320** and provide the third control signal BTS_CHG to the second power MOSFET **320** over a pre-defined time period when a control signal indicating the asynchronous mode or the pulse skip mode is applied to the cyclic bootstrap charger **350**. The third control signal BTS_CHG may be different from the first control signal PWM_HS and the second control signal PWM_LS.

(40) FIG. 4 is a timing diagram illustrating the pulse skip mode according to an example embodiment of the inventive concept.

(41) Referring to FIG. 4, the PWM controller **340** and the cyclic bootstrap charger **350** may enter the pulse skip mode (PSK high) at a first time point T1. For example, the PWM controller **340** and the cyclic bootstrap charger **350** may receive the mode signal SIG_MODE indicating the pulse skip mode at the first time point T1.

(42) The PWM controller **340** may inactivate the first control signal PWM_HS and the second control signal PWM_LS in response to the mode signal SIG_MODE indicating the pulse skip mode. This is because the PWM controller **340** does not switch the first power MOSFET **310** and the second power MOSFET **320** in the pulse skip mode. In this case, at the first time point T1, the second control signal PWM_LS may have a low level.

(43) The cyclic bootstrap charger **350** may periodically generate the third control signal BTS_CHG for forcibly turning on the second power MOSFET **320**, in response to the mode signal SIG_MODE indicating the pulse skip mode. The second gate drive signal GD_LS may be toggled according to the third control signal BTS_CHG which is periodically generated even though the second control signal PWM_LS is logic low based on the pulse skip mode. The cyclic bootstrap charger **350** may toggle the third control signal BTS_CHG by a pre-defined number of times every first cycle t.sub.CYC. A period in which the third control signal BTS_CHG is toggled by the pre-defined number of times may be referred as a charge period t.sub.CHG.

(44) According to an embodiment of the inventive concept, when the cyclic bootstrap charger **350** starts to toggle the third control signal BTS_CHG at a second time point T2, the third control signal BTS_CHG may be toggled four times, and the third control signal BTS_CHG may be toggled again at a third time point T3 at which time elapses by the first cycle t.sub.CYC from the second time point T2. In other words, the third control signal BTS_CHG may be toggled again after the first cycle t.sub.CYC completes. According to an embodiment of the inventive concept, the third control signal BTS_CHG may be toggled in a second cycle (also referred to as switching time) t.sub.SW. For example, when the third control signal BTS_CHG transitions to “logic high” at the second time point T2, the third control signal BTS_CHG may transition to “logic high” again at a fourth time point T4. In other words, the second cycle t.sub.SW may correspond to a time interval between the second time point T2 and the fourth time point T4. According to an embodiment of the inventive concept, the third control signal BTS_CHG may be toggled by the pre-defined number of times and then maintained “logic low” for a discharge period t.sub.DISCHG. During the discharge period t.sub.DISCHG, the third control signal BTS_CHG is not applied, and thus, the voltage level VBTS of the bootstrap capacitor **330** may slowly decrease.

(45) According to various embodiments of the inventive concept, the first cycle t.sub.CYC, the

second cycle $t_{sub.SW}$, and the discharge period $t_{sub.DISCHG}$ may be variably controlled based on a capacitance value of the bootstrap capacitor **330** and a leakage current magnitude of the bootstrap capacitor **330**. For example, the larger a capacity of the bootstrap capacitor **330**, the greater a length of the discharge period $t_{sub.DISCHG}$. As the capacity of the bootstrap capacitor **330** increases, the number of toggles of the third control signal BTS_CHG may increase. As another example, the larger the leakage current of the bootstrap capacitor **330**, the less the length of the discharge period $t_{sub.DISCHG}$. As the leakage current of the bootstrap capacitor **330** increases, the number of toggles of the third control signal BTS_CHG may increase.

(46) FIG. 5 is a timing diagram illustrating the asynchronous mode according to an embodiment of the inventive concept.

(47) Referring to FIG. 5, the PWM controller **340** and the cyclic bootstrap charger **350** may enter the asynchronous mode (Async high) at a fifth time point $T5$. For example, the PWM controller **340** and the cyclic bootstrap charger **350** may receive the mode signal SIG_MODE indicating the asynchronous mode at the fifth time point $T5$.

(48) The PWM controller **340** may inactivate the second control signal PWM_LS and toggle only the first control signal PWM_HS in response to the mode signal SIG_MODE indicating the asynchronous mode. The cyclic bootstrap charger **350** may toggle the third control signal BTS_CHG in response to the mode signal SIG_MODE indicating the asynchronous mode. The second gate drive signal GD_LS may be toggled according to the third control signal BTS_CHG being toggled based on the asynchronous mode even though the second control signal PWM_LS remains logic low based on the asynchronous mode. According to an embodiment of the inventive concept, the third control signal BTS_CHG may be activated alternately with the first control signal PWM_HS . In other words, the third control signal BTS_CHG may be activated only in a period in which the first control signal PWM_HS is “logic low” during toggling of the first control signal PWM_HS . In other words, the third control signal BTS_CHG may be “logic high” when the first control signal PWM_HS is “logic low” during toggling of the first control signal PWM_HS .

(49) Referring to FIG. 5, the first control signal PWM_HS may be toggled at a sixth time point $T6$. Periods in which the first control signal PWM_HS is turned on and off may be determined according to a pre-defined PWM ratio. Hereinafter, for convenience of description, it may be assumed that a ratio of a turning-on period to a turning-off period is 1:1 and the first control signal PWM_HS is toggled at every switching time $t_{sub.SW}$.

(50) When the switching time $t_{sub.SW}$ elapses after the first control signal PWM_HS is toggled at the sixth time point $T6$, the third control signal BTS_CHG may be activated. A length of a time for which the third control signal BTS_CHG is activated may be less than or equal to a length of the switching time $t_{sub.SW}$. This is because a length of a time for which the first control signal PWM_HS is inactivated is the same as the length of the switching time $t_{sub.SW}$. If it is assumed that a time point at which the third control signal BTS_CHG starts to be toggled is a seventh time point $T7$, the seventh time point $T7$ may be a time point which elapses by the switching time $t_{sub.SW}$ from the sixth time point $T6$.

(51) The third control signal BTS_CHG may be toggled by the pre-defined number of times and then maintained “logic low” for the discharge period $t_{sub.DISCHG}$. Because the voltage level $VBTS$ of the bootstrap capacitor **330** is maintained in a proper level according to the toggling of the third control signal BTS_CHG , the toggling of the third control signal BTS_CHG causing noise in a waveform of an inductor current I_{IND} does not have to be continuously maintained.

(52) The charge period $t_{sub.CHG}$ of the bootstrap capacitor **330** may be a time taken to toggle the third control signal BTS_CHG by the pre-defined number of times. Referring to FIG. 5, the charge period $t_{sub.CHG}$ may include an eighth time point $T8$ at which the third control signal BTS_CHG transitions to “logic low” after being toggled four times since the seventh time point $T7$. In addition, the discharge period $t_{sub.DISCHG}$ may include a ninth time point $T9$ at which the third control signal BTS_CHG starts to be toggled again since the eighth time point $T8$. A charging cycle

t.sub.CYC of the bootstrap capacitor **330** may include the charge period t.sub.CHG and the discharge period t.sub.DISCHG.

(53) According to various embodiments of the inventive concept, the charge period t.sub.CHG and the discharge period t.sub.DISCHG may be variably controlled based on the capacitance value of the bootstrap capacitor **330** and the leakage current magnitude of the bootstrap capacitor **330**. For example, the larger the capacity of the bootstrap capacitor **330**, the greater the length of the discharge period t.sub.DISCHG. As the capacity of the bootstrap capacitor **330** increases, the number of toggles of the third control signal BTS_CHG may increase, thereby increasing the charge period t.sub.CHG. As another example, as the leakage current of the bootstrap capacitor **330** increases, the voltage level VBTS of the bootstrap capacitor **330** may quickly drop, thereby decreasing the length of the discharge period t.sub.DISCHG. As the leakage current of the bootstrap capacitor **330** increases, an amount of charges discharged while toggling the third control signal BTS_CHG may increase. Therefore, as the leakage current of the bootstrap capacitor **330** increases, the number of toggles of the third control signal BTS_CHG may increase, thereby increasing a length of the charge period t.sub.CHG. In an example embodiment, the length of the discharge period t.sub.DISCHG is determined in proportion to the capacitance value of the bootstrap capacitor **330**.

(54) FIG. **6** is a circuit diagram illustrating the buck converter **300** including the bootstrap capacitor **330**, according to a comparative example.

(55) Referring to FIG. **6**, the buck converter **300** may not include the cyclic bootstrap charger **350**. In other words, the buck converter **300** does not generate the third control signal BTS_CHG for turning on the second power MOSFET **320** when receiving the mode signal SIG_MODE indicates the asynchronous mode or the pulse skip mode.

(56) The buck converter **300** of FIG. **6** may further include a bootstrap detection circuit **370**. The bootstrap detection circuit **370** may monitor the voltage level VBTS of the bootstrap capacitor **330** and generate a control signal BTS_UVLO by comparing the voltage level VBTS of the bootstrap capacitor **330** with a threshold level. When the PWM controller **340** receives the control signal BTS_UVLO from the bootstrap detection circuit **370**, the PWM controller **340** may determine there is a situation in which the voltage level VBTS of the bootstrap capacitor **330** is lower than the threshold level. The PWM controller **340** may perform toggling of the second control signal PWM_LS to charge the bootstrap capacitor **330**.

(57) However, the reception of the control signal BTS_UVLO by the PWM controller **340** from the bootstrap detection circuit **370** indicates that the voltage level VBTS of the bootstrap capacitor **330** is already too low, and due to the already too low voltage level of the bootstrap capacitor **330**, a protection operation on the charging IC **110** is performed.

(58) FIGS. **7A** and **7B** are timing diagrams illustrating the pulse skip mode according to comparative examples.

(59) FIG. **7A** shows a case where a period for operating in the pulse skip mode is long, and FIG. **7B** shows a case where a period for operating in the pulse skip mode is short.

(60) Referring to FIG. **7A**, a first period PERIOD **1** may correspond to an operation of the buck converter **300** according to a comparative example. After entering the pulse skip mode, the second control signal PWM_LS for charging the bootstrap capacitor **330** may transition to “logic high” in the first period PERIOD **1**. In the first period PERIOD **1**, the second control signal PWM_LS may maintain a “logic high” state for a pre-defined charge period t.sub.CHG instead of being toggled. When the second control signal PWM_LS maintains being “logic high” for the charge period t.sub.CHG, a peak value of the inductor current I_IND may be generated to be large.

(61) A second period PERIOD **2** may correspond to an operation of the buck converter **300** according to an example embodiment of the inventive concept. In other words, in the second period PERIOD **2**, the second control signal PWM_LS may be toggled while the second power MOSFET **320** is being repeatedly turned-on and off every switching time t.sub.SW. For example, a total sum

of activation times of the second control signal PWM_LS according to toggling four times may be the same as a time of the charge period $t_{sub.CHG}$ of the second control signal PWM_LS in the first period PERIOD 1. Thus, even when the second control signal PWM_LS is applied for the same time, the peak value of the inductor current I_{IND} may be reduced by controlling the second power MOSFET 320 based on shortly repeated toggle signals.

(62) Referring to FIGS. 6 and 7B, the buck converter 300 according to a comparative example may monitor the voltage level of the bootstrap capacitor 330. For example, the PWM controller 340 may receive the control signal BTS_UVLO from the bootstrap detection circuit 370 at a tenth time point T10. The PWM controller 340 may activate the second control signal PWM_LS for charging the bootstrap capacitor 330, in response to the reception of the control signal BTS_UVLO.

Therefore, the PWM controller 340 may inactivate the pulse skip mode for a period of time and toggle the second control signal PWM_LS. In other words, the buck converter 300 according to a comparative example periodically inactivates the pulse skip mode and applies the second control signal PWM_LS to charge the bootstrap capacitor 330, and thus, a magnitude of a ripple component of an output voltage VOUT may be large.

(63) FIG. 8 is a timing diagram illustrating the asynchronous mode according to a comparative example.

(64) Referring to FIG. 8, the buck converter 300 according to a comparative example may monitor the voltage level VBTS of the bootstrap capacitor 330 based on the bootstrap detection circuit 370. For example, the PWM controller 340 may receive the control signal BTS_UVLO from the bootstrap detection circuit 370 at an eleventh time point T11. The PWM controller 340 may activate the second control signal PWM_LS for charging the bootstrap capacitor 330, in response to the reception of the control signal BTS_UVLO. Because the asynchronous mode being activated at the eleventh time point T11 is a mode of inactivating the second control signal PWM_LS, the PWM controller 340 may inactivate the asynchronous mode for a period of time and toggle the second control signal PWM_LS. In other words, the buck converter 300 according to a comparative example periodically inactivates the asynchronous mode and applies the second control signal PWM_LS to charge the bootstrap capacitor 330, and thus, a magnitude of a ripple component of the output voltage VOUT may be large.

(65) While the inventive concept has been particularly shown and described with reference to embodiments thereof, it will be understood that various changes in form and details may be made thereto without departing from the spirit and scope of the inventive concept set forth by the following claims.

Claims

1. A buck converter, comprising: a first switching element connecting an input voltage node to a switching node; a second switching element connecting the switching node to a ground node; a low drop out (LDO) voltage regulator connected to the input voltage node; a pulse width modulation (PWM) controller configured to receive a mode signal indicating an operation mode and toggle, in response to the mode signal, a first control signal for turning on the first switching element or a second control signal for turning on the second switching element; a bootstrap charger configured to receive the mode signal and repeatedly switch levels of a third control signal for turning on the second switching element between a high state and a low state while the first control signal and the second control signal are in the low state in a pulse skip mode, and repeatedly switch the levels of the third control signal between the high state and the low state while levels of the first control signal are repeatedly switched between the low state and the high state, respectively, in an asynchronous mode; and a bootstrap capacitor connecting an output node of the LDO voltage regulator to the switching node.

2. The buck converter of claim 1, wherein the operation mode corresponds to the pulse skip mode,

the PWM controller is further configured to inactivate the first control signal and the second control signal in the pulse skip mode, and the bootstrap charger is further configured to toggle the third control signal.

3. The buck converter of claim 2, wherein the third control signal is toggled a pre-defined number of times in a charge period and maintains a logic low level in a discharge period after the charge period.

4. The buck converter of claim 3, wherein a length of the charge period, a length of the discharge period, and the pre-defined number of times are determined based on a magnitude of a leakage current of the bootstrap capacitor and a capacitance value of the bootstrap capacitor.

5. The buck converter of claim 1, wherein the operation mode corresponds to the asynchronous mode, the PWM controller is further configured to inactivate the second control signal in the asynchronous mode.

6. The buck converter of claim 5, wherein the third control signal is toggled a pre-defined number of times in a charge period and maintains a logic low level in a discharge period after the charge period.

7. The buck converter of claim 6, wherein a length of the charge period and a length of the discharge period are determined based on a magnitude of a leakage current of the bootstrap capacitor and a capacitance value of the bootstrap capacitor.

8. The buck converter of claim 7, wherein the length of the charge period is directly proportional to the magnitude of the leakage current of the bootstrap capacitor, and the length of the discharge period is determined in proportion to the capacitance value of the bootstrap capacitor.

9. An operating method of a buck converter, the operating method comprising: receiving a mode signal and identifying an operation mode; toggling, based on the operation mode, a first control signal for turning on a first switching element or a second control signal for turning on a second switching element; and repeatedly switch levels of a third control signal for turning on the second switching element between a high state and a low state while the first control signal and the second control signal are in the low state in a pulse skip mode, and repeatedly switch the levels of the third control signal between the high state and the low state while levels of the first control signal are repeatedly switched between the low state and the high state, respectively, in an asynchronous mode.

10. The operating method of claim 9, further comprising, when the operation mode corresponds to the pulse skip mode, bypassing the toggling of the first control signal and the second control signal.

11. The operating method of claim 10, wherein the third control signal is toggled a pre-defined number of times in a charge period and inactivated in a discharge period after the charge period.

12. The operating method of claim 11, wherein a length of the charge period, a length of the discharge period, and the pre-defined number of times are determined based on a magnitude of a leakage current of a bootstrap capacitor included in the buck converter and a capacitance value of the bootstrap capacitor.

13. The operating method of claim 9, further comprising, when the operation mode corresponds to the asynchronous mode, inactivating the second control signal.

14. The operating method of claim 13, wherein the third control signal is toggled a pre-defined number of times in a charge period and inactivated in a discharge period after the charge period.

15. The operating method of claim 14, wherein a length of the charge period and a length of the discharge period are determined based on a magnitude of a leakage current of a bootstrap capacitor included in the buck converter and a capacitance value of the bootstrap capacitor.

16. The operating method of claim 15, wherein the length of the charge period is inversely proportional to the magnitude of the leakage current of the bootstrap capacitor, and the length of the discharge period is determined in proportion to the capacitance value of the bootstrap capacitor.

17. A buck converter, comprising: a first switching element connecting an input voltage node to a switching node; a second switching element connecting the switching node to a ground node; a

pulse width modulation (PWM) controller configured to receive a mode signal indicating an operation mode and toggle, in response to the mode signal, a first control signal for turning on the first switching element or a second control signal for turning on the second switching element; a bootstrap charger configured to receive the mode signal and repeatedly switch levels of a third control signal for turning on the second switching element between a high state and a low state while the first control signal and the second control signal are in the low state in a pulse skip mode, and repeatedly switch the levels of the third control signal between the high state and the low state while levels of the first control signal are repeatedly switched between the low state and the high state, respectively, in an asynchronous mode; and a bootstrap capacitor configured to boost a voltage input to a gate of the first switching element, wherein the bootstrap capacitor is connected to the switching node.

18. The buck converter of claim 17, wherein, when the operation mode corresponds to the pulse skip mode, the PWM controller is further configured to inactivate the first control signal and the second control signal, and the bootstrap charger is further configured to toggle the third control signal, and when the operation mode corresponds to the asynchronous mode, the PWM controller is further configured to inactivate the second control signal.
